

尝试推导无松弛变量的系统

二次规划问题

$$\begin{aligned} \min_x \quad & f(x) = \frac{1}{2}x^\top Px + c^\top x \\ \text{s. t.} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

记 $\xi = h - Gx$ ，并记 $H(x) = Ax - b$

注意这里的 ξ 只是 x 的函数，不作为松弛变量。

改写了 $F(x, s; \mu, \rho)$ 的定义（代入函数 Γ （IPRM 中的函数 G ）的值从 x_j 改为了 ξ_j ）：

$$F(x, s; \mu, \rho) = f(x) + \sum_{j=1}^n \Gamma(\xi_j, s_j; \mu, \rho)$$

Theorem 2.5 中子问题（17）关联的系统（18）-（20）变为

$$\begin{aligned} \nabla f(x) - \nabla H(x)\lambda + \rho G^\top y &= 0 \\ H(x) &= 0 \\ z - \xi &= 0 \end{aligned}$$

由 Lemma 2.1（1）， $z - \xi = 0$ 蕴含 $\rho y = s$

代入 $\xi = h - Gx$ ，并且令 $\lambda = -t$ ，则系统变为：

$$\begin{aligned} Px + c + A^\top t + G^\top s &= 0 \\ Ax &= b \\ Gx + z &= h \end{aligned}$$

因为之前推导时用 ξ 代替 x 代入 z, y ，所以这里

$$\begin{aligned} z_j &= z_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho} \left(\sqrt{(s_j - \rho\xi_j)^2 + 4\rho\mu} - (s_j - \rho\xi_j) \right) \\ y_j &= y_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho} \left(\sqrt{(s_j - \rho\xi_j)^2 + 4\rho\mu} + (s_j - \rho\xi_j) \right) \end{aligned}$$

系统以 μ, x, t, s 为迭代变量。

定义

$$\phi_{(\mu, \rho)}(x, t, s) = \frac{1}{2} \|\psi_{(\mu, \rho)}(x, t, s)\|^2 \quad (1)$$

$$\psi_{(\mu, \rho)}(x, t, s) = \begin{bmatrix} Px + c + A^\top t + G^\top s \\ Ax - b \\ Gx + z - h \end{bmatrix} \quad (2)$$

完整的目标为

$$\mu + \gamma \phi_{(\mu, \rho)}(x, t, s) = 0 \quad (3)$$

$$\psi_{(\mu, \rho)}(x, t, s) = 0 \quad (4)$$

(3)的牛顿方程为

$$\gamma \nabla_{(\mu, x, t, s)} \phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k)^\top \begin{bmatrix} \Delta\mu \\ d_x \\ d_t \\ d_s \end{bmatrix} + \Delta\mu = -\mu_k - \gamma \phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) \quad (5)$$

(4)的牛顿方程为

$$\nabla_{(\mu, x, t, s)} \psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k)^\top \begin{bmatrix} \Delta\mu \\ d_x \\ d_t \\ d_s \end{bmatrix} = -\psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) \quad (6)$$

对(1)求导得

$$\nabla_{(\mu, x, t, s)} \phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) = \nabla_{(\mu, x, \lambda, s)} \psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) \psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) \quad (7)$$

将(7)(6)依次代入(5)得

$$-2\gamma \phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) + \Delta\mu = -\mu_k - \gamma \phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k)$$

得到

$$\Delta\mu = -\mu_k + \gamma \phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) \quad (8)$$

$$\psi_{(\mu, \rho)}(x, t, s) = \begin{bmatrix} Px + c + A^\top t + G^\top s \\ Ax - b \\ Gx + z - h \end{bmatrix}$$

$$\nabla_{(\mu, x, t, s)} \psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k) = \begin{bmatrix} 0 & 0 & \frac{1}{\rho_k} e^\top (Z_k + Y_k)^{-1} \\ P & A^\top & G^\top Y_k (Z_k + Y_k)^{-1} \\ A & 0 & 0 \\ G & 0 & -\frac{1}{\rho_k} Z_k (Z_k + Y_k)^{-1} \end{bmatrix}$$

代入(6):

$$\begin{bmatrix} 0 & P & A^\top & G^\top \\ 0 & A & 0 & 0 \\ \frac{1}{\rho_k} (Z_k + Y_k)^{-1} e & Y_k (Z_k + Y_k)^{-1} G & 0 & -\frac{1}{\rho_k} (Z_k + Y_k)^{-1} Z_k \end{bmatrix} \begin{bmatrix} \Delta\mu \\ d_x \\ d_t \\ d_s \end{bmatrix} = \begin{bmatrix} -r_k^d \\ -r_k^h \\ -r_k^e \end{bmatrix}$$

$$\begin{bmatrix} P & A^\top & G^\top \\ A & O & O \\ Y_k(Z_k + Y_k)^{-1}G & O & -\frac{1}{\rho_k}(Z_k + Y_k)^{-1}Z_k \end{bmatrix} \begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = \begin{bmatrix} -r_k^d \\ -r_k^h \\ -r_k^e - \frac{1}{\rho_k}\Delta\mu(Z_k + Y_k)^{-1}e \end{bmatrix}$$

$$\begin{bmatrix} P & A^\top & G^\top \\ A & O & O \\ G & O & -\frac{1}{\rho_k}Y_k^{-1}Z_k \end{bmatrix} \begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = - \begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k + Y_k)Y_k^{-1}r_k^e + \frac{1}{\rho_k}\Delta\mu Y_k^{-1}e \end{bmatrix}$$

$$r_k^d = Px_k + c + A^\top t_k + G^\top s_k$$

$$r_k^h = Ax_k - b$$

$$r_k^e = Gx_k + z_k - h$$

推导一下与 IPRM 的 MATLAB 实现一致的方程组形式：

用第三行乘以 $\rho_k G^\top Z_k^{-1} Y_k$ 加到第一行，消去第一行的 $G^\top$

$$\begin{bmatrix} P + \rho_k G^\top Z_k^{-1} Y_k G & A^\top & O \\ A & O & O \\ G & O & -\frac{1}{\rho_k} Y_k^{-1} Z_k \end{bmatrix} \begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = - \begin{bmatrix} r_k^d + \rho_k G^\top Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta\mu G^\top Z_k^{-1} e \\ r_k^h \\ (Z_k + Y_k) Y_k^{-1} r_k^e + \frac{1}{\rho_k} \Delta\mu Y_k^{-1} e \end{bmatrix}$$

提取出第三行：

$$\begin{bmatrix} P + \rho_k G^\top Z_k^{-1} Y_k G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} d_x \\ d_t \end{bmatrix} = - \begin{bmatrix} r_k^d + \rho_k G^\top Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta\mu G^\top Z_k^{-1} e \\ r_k^h \end{bmatrix}$$

$$Gd_x - \frac{1}{\rho_k} Y_k^{-1} Z_k d_s + (Z_k + Y_k) Y_k^{-1} r_k^e + \frac{1}{\rho_k} \Delta\mu Y_k^{-1} e = 0$$

$$Gd_x + (Z_k + Y_k) Y_k^{-1} r_k^e + \frac{1}{\rho_k} \Delta\mu Y_k^{-1} e = \frac{1}{\rho_k} Y_k^{-1} Z_k d_s$$

$$d_s = \rho_k Z_k^{-1} Y_k G d_x + \rho_k Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta\mu Z_k^{-1} e$$

$$\text{代入 } Z_k^{-1} = \frac{\rho_k}{\mu_k} Y_k$$

$$\begin{bmatrix} P + \frac{\rho_k^2}{\mu_k} G^\top Y_k^2 G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} d_x \\ d_t \end{bmatrix} = - \begin{bmatrix} r_k^d + \rho_k G^\top Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta \mu G^\top Z_k^{-1} e \\ r_k^h \end{bmatrix}$$

$$d_s = \frac{\rho_k^2}{\mu_k} Y_k^2 G d_x + \frac{\rho_k^2}{\mu_k} Y_k (Z_k + Y_k) r_k^e + \frac{\Delta \mu \rho_k}{\mu_k} y_k$$

$$d_s = \frac{\rho_k^2}{\mu_k} Y_k^2 (G d_x + r_k^e) + \rho_k r_k^e + \frac{\Delta \mu \rho_k}{\mu_k} y_k$$